Differentiable explicit impact model of visco-hyperelastic beam lattices with frictional contact

Physics, contact, material, element, time integration, test rig and exact sensitivities

Model documentation

code: `src/explicitFE.py`, `explicitSolve.py`, `contact.py`, `latticeSetup.py`, `material.py`, `adjoint.py`, `objectives.py`, `optimizeDrop.py`

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Abstract

This document describes the current simulation model used to compute the impact response of additively manufactured beam lattices and its exact design sensitivities. A lattice of slender struts made of an elastomeric polyurethane (Carbon EPU 45) is bonded to a rigid base plate and capped by a rigid top plate; a rigid impactor carrying a prescribed kinetic energy loads the top plate. The struts are modelled with a corotational three-dimensional Timoshenko beam element whose section law is visco-hyperelastic (Yeoh hyperelasticity integrated over the cross-section, quasi-linear Prony viscoelasticity). Contact between struts is represented by a line-contact density integrated along the beams, contact with both plates by Hertzian node contact, and the impactor–plate interaction by a critically damped Hertz law; regularised Coulomb friction acts at every contact. The equations of motion are integrated with an explicit central-difference (velocity Verlet) scheme with an exact exponential-map rotation update and an adaptive stable time step. Design gradients are obtained with a checkpointed discrete adjoint of the time-stepping scheme. All model components are written in PyTorch, so every force is differentiable with respect to positions, rotations and section properties.

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1 Units, notation and overview

All quantities use the consistent set **mm, N, s, tonne (t), MPa**: $1\,\mathrm{t\,mm\,s^{-2}} = 1\,\mathrm{N}$, $1\,\mathrm{J} = 1000\,\mathrm{N\,mm}$ and a density of $1\,\mathrm{kg\,m^{-3}} = 10^{-12}\,\mathrm{t\,mm^{-3}}$. Vectors are bold, $[\,\boldsymbol{a}\,]_\times$ is the skew matrix with $[\,\boldsymbol{a}\,]_\times \boldsymbol{b} = \boldsymbol{a}\times\boldsymbol{b}$, and $\exp([\,\boldsymbol{\theta}\,]_\times)$ is the rotation by the rotation vector $\boldsymbol{\theta}$. $\mathrm{softplus}(s) = \ln(1+e^s)$, $\sigma(s) = 1/(1+e^{-s})$ and $\mathrm{relu}(s) = \max(s, 0)$.

The model has the following parts, described in turn:

1. material (Section 2): EPU 45 visco-hyperelastic constants;
2. lattice geometry, sections and design variables (Section 3);
3. corotational Timoshenko beam element (Section 4);
4. section law: fibre-integrated Yeoh hyperelasticity with Prony viscoelasticity (Section 5);
5. inertia (Section 6);
6. test rig and boundary conditions: bonded base plate, tied top plate, impactor (Section 7);
7. contact: impactor–plate, beam–beam, beam–plate, friction (Section 8);
8. explicit time integration and the stable time step (Section 9);
9. energy accounting (Section 10);
10. response measures, objectives and exact sensitivities (Section 11);
11. verification (Section 12) and limitations (Section 13).

2 Material: Carbon EPU 45

EPU 45 is an elastomeric polyurethane printed by digital light synthesis (DLS). Its response is strongly rate-dependent and dissipative. The model uses a hyperelastic long-term response with a single relaxation term.

2.1 Constants

quantity	value	note
density ρ	$1.06\times10^{-9}\,\mathrm{t\,mm^{-3}}$ $(1060\,\mathrm{kg\,m^{-3}})$	Carbon TDS (ASTM D792)
Yeoh C_{10}, C_{20}, C_{30}	2.4, 0.15, 0.02 MPa	long-term, compression-calibrated
compressibility D_i	≈ 0	nearly incompressible
Prony g_1, k_1	0.958, 0	shear relaxation only
Prony τ_1	$5.2\times10^{-5}\,\mathrm{s}$	
Poisson ratio ν	0.49	assumed
long-term modulus $E_\infty = 6C_{10}$	14.4 MPa	small-strain limit of Yeoh
instantaneous modulus $E_0 = E_\infty/(1-g_1)$	342.9 MPa	governs the explicit time step
shear moduli G_∞, G_0	$E/(2(1+\nu))$	

Table 1: EPU 45 constants (`config/materials.yaml`, entry `EPU_45`; `material.epu45()`).

E_∞ agrees with the manufacturer's tensile modulus (17 MPa at 500 mm/min, i.e. close to the relaxed state). The Yeoh constants are calibrated in compression; they are stiffer than the tensile

data sheet at large tensile strain (5.5/11.3/40.8 MPa vs. 3/4/9 MPa at 50/100/200 % strain), which is the expected tension/compression asymmetry of such fits. With one Prony term the storage and loss moduli are

$$E'(\omega) = E_\infty + (E_0 - E_\infty)\frac{\omega^2\tau_1^2}{1+\omega^2\tau_1^2}, \qquad E''(\omega) = (E_0 - E_\infty)\frac{\omega\tau_1}{1+\omega^2\tau_1^2}, \tag{1}$$

giving $\tan\delta = E''/E' = 0.73$ at 100 Hz and 2.1 at 1 kHz, i.e. strong damping at impact frequencies. (At 1 Hz the model gives $E' = 14.4$ MPa, whereas the manufacturer's DMA reports ≈ 65 MPa near $T_g = 30\,^\circ$C; the slow viscoelasticity close to T_g is not represented.)

2.2 Yeoh hyperelasticity

For an incompressible Yeoh solid the strain-energy density is

$$W(\bar{I}_1) = C_{10}(\bar{I}_1 - 3) + C_{20}(\bar{I}_1 - 3)^2 + C_{30}(\bar{I}_1 - 3)^3. \tag{2}$$

For uniaxial stretch λ, $\bar{I}_1 - 3 = \lambda^2 + 2/\lambda - 3$ and the nominal (first Piola) stress is

$$P(\lambda) = 2\left(\lambda - \lambda^{-2}\right)\left[C_{10} + 2C_{20}(\bar{I}_1 - 3) + 3C_{30}(\bar{I}_1 - 3)^2\right], \qquad \left.\frac{\mathrm{d}P}{\mathrm{d}\lambda}\right|_{\lambda=1} = 6C_{10} = E_\infty. \tag{3}$$

$P \to -\infty$ as $\lambda \to 0$, so a strut element cannot be compressed to zero length. For simple shear γ, $\bar{I}_1 - 3 = \gamma^2$ and $\tau = 2\gamma\left[C_{10} + 2C_{20}\gamma^2 + 3C_{30}\gamma^4\right]$.

3 Lattice, cross-sections and design variables

3.1 Lattice

The lattice is the C4 cubic cell of the original optimisation study (exampleCase 3), $3 \times 3 \times 1$ cells of size 12.7 mm, i.e. a $38.1 \times 38.1 \times 12.7$ mm coupon with 168 nodes and 276 struts (strut lengths 2.6–6.4 mm). Each strut is subdivided into elements of target length h_e (number of elements per strut = round(L/h_e)); $h_e = 1$ mm (1332 elements) is the converged results mesh, $h_e = 3$ mm (444 elements) is used for fast tests. The mapping element→strut is kept (`beam_of_elem`).

3.2 Circular sections

Each element has a solid circular section of radius r:

$$A = \pi r^2, \quad A_s = \kappa A, \quad I_y = I_z = \tfrac{\pi}{4}r^4, \quad J = I_y + I_z, \qquad \kappa = \frac{6(1+\nu)}{7+6\nu} = 0.899 \text{ (Cowper)}. \tag{4}$$

Struts are short and stout ($L/r = 3$–8), so the shear coefficient matters.

3.3 Design variables and symmetry

Design rows $\boldsymbol{x} \in [0,1]^n$ are mapped to radii by $r = r_{\min} + (r_{\max} - r_{\min})\,x$ inside the objective (e.g. $[r_{\min}, r_{\max}] = [0.4, 1.2]$ mm), then expanded by a symmetry rule (*Independent*: one row per strut; *Unitcell*: one row per strut of the unit cell, repeated; *OneValue*: a single radius; *Midplane*: three groups by height), and finally to every element of each strut. The expansion is linear, so its derivative is exact.

4 Corotational Timoshenko beam element

4.1 Kinematic state

Each node carries its position $\boldsymbol{x}_a = \boldsymbol{X}_a + \boldsymbol{u}_a$ and a rotation $\mathbf{R}_a \in SO(3)$ measured from the reference configuration. No element-level state is stored.

For element $e = (1, 2)$ with reference axes $\mathbf{T}^0 = [\boldsymbol{e}_1^0, \boldsymbol{e}_2^0, \boldsymbol{e}_3^0]^\top$ (axis 1 along the chord; axis 2 from the global axis least aligned with the chord), the current nodal triads are $\mathbf{t}_a = \mathbf{R}_a \mathbf{T}^{0\top}$ (columns $\boldsymbol{n}_x^a, \boldsymbol{n}_y^a, \boldsymbol{n}_z^a$).

4.2 Element frame

$$L = \|\boldsymbol{x}_2 - \boldsymbol{x}_1\|, \quad \boldsymbol{e}_1 = \frac{\boldsymbol{x}_2 - \boldsymbol{x}_1}{L}, \quad \boldsymbol{q} = \tfrac{1}{2}\left(\boldsymbol{n}_y^1 + \boldsymbol{n}_y^2\right), \quad \boldsymbol{e}_3 = \frac{\boldsymbol{e}_1 \times \boldsymbol{q}}{\|\boldsymbol{e}_1 \times \boldsymbol{q}\|}, \quad \boldsymbol{e}_2 = \boldsymbol{e}_3 \times \boldsymbol{e}_1. \tag{5}$$

This frame follows Battini and Pacoste [1] and needs no stored state.

4.3 Local deformations

The rotation of node a relative to the element frame, $\mathbf{Q}_a = \mathbf{E}^\top \mathbf{t}_a$ with $\mathbf{E} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3]$, is converted to a rotation vector by the logarithm

$$\boldsymbol{a} = \tfrac{1}{2}\,\mathrm{vec}\left(\mathbf{Q}_a - \mathbf{Q}_a^\top\right), \quad s = \|\boldsymbol{a}\| = \sin\phi, \quad c = \tfrac{1}{2}(\mathrm{tr}\,\mathbf{Q}_a - 1) = \cos\phi, \quad \boldsymbol{\theta}_a = \boldsymbol{a}\,\frac{\mathrm{atan2}(s, c)}{s}. \tag{6}$$

This form is smooth for $\phi \in [0, \pi)$. (The arcsine form used in the earlier linear code stiffens like $1/\cos\phi$ and diverges at 90° element-relative rotation, which occurs during crushing.) The six local deformations are

$$\boldsymbol{d} = (\Delta L,\ r_x,\ r_{y1},\ r_{z1},\ r_{y2},\ r_{z2}), \quad \Delta L = L - L_0, \quad r_x = \theta_x^2 - \theta_x^1, \quad r_{ya} = \theta_y^a,\ r_{za} = \theta_z^a. \tag{7}$$

4.4 Internal force by virtual work

With local resultants $\boldsymbol{f}_{\mathrm{loc}}(\boldsymbol{d}) = (N, T, M_{y1}, M_{z1}, M_{y2}, M_{z2})$ from the section law (Section 5), the global nodal forces and moments are the exact virtual-work transpose

$$\begin{bmatrix} \boldsymbol{f}_u \\ \boldsymbol{f}_\theta \end{bmatrix} = \left(\frac{\partial \boldsymbol{d}}{\partial(\boldsymbol{u}, \delta\boldsymbol{\theta})}\right)^\top \boldsymbol{f}_{\mathrm{loc}}, \qquad \mathbf{R}_a \leftarrow (\mathbf{I} + [\,\delta\boldsymbol{\theta}_a\,]_\times)\,\mathbf{R}_a, \tag{8}$$

i.e. a vector–Jacobian product computed by automatic differentiation, with rotational variations taken as spatial spins. For an elastic law, $\boldsymbol{f}_{\mathrm{loc}} = \partial\Pi/\partial\boldsymbol{d}$ and (8) is the exact gradient of the strain energy, so the explicit dynamics conserves energy. The geometric (P–δ) stiffness is included automatically.

4.5 Linear Timoshenko resultants

The small-strain part of the section law is the exact two-node Timoshenko element (per bending plane, with $\Omega = EI/(GA_sL_0^2)$, $\mu = 1 + 12\Omega$, $\lambda = 1 + 3\Omega$, $\gamma = 1 - 6\Omega$):

$$N = \frac{EA}{L_0}\Delta L, \quad T = \frac{GJ}{L_0} r_x, \quad \begin{bmatrix} M_1 \\ M_2 \end{bmatrix} = \frac{2EI}{\mu L_0} \begin{bmatrix} 2\lambda & \gamma \\ \gamma & 2\lambda \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}. \tag{9}$$

Its small-displacement tangent equals the classical 12×12 Timoshenko stiffness to 10^{-9}.

5 Section law: fibre-integrated Yeoh with Prony viscoelasticity

5.1 Large-strain long-term response

Axial stretch, bending and torsion all use the Yeoh law, integrated over the cross-section. From the local deformations the element strains are

$$\varepsilon_0 = \frac{\Delta L}{L_0}, \quad \kappa_y = \frac{r_{y2} - r_{y1}}{L_0}, \quad \kappa_z = \frac{r_{z2} - r_{z1}}{L_0}, \quad \vartheta = \frac{r_x}{L_0}, \tag{10}$$

and a fibre at section coordinates (y, z) has stretch and torsional shear

$$\lambda_f = 1 + \varepsilon_0 - y\kappa_z + z\kappa_y, \qquad \gamma_f = \vartheta\sqrt{y^2 + z^2}, \qquad \bar{I}_1 - 3 = \lambda_f^2 + 2/\lambda_f - 3 + \gamma_f^2. \tag{11}$$

The element energy is

$$U_{\text{fib}}(\boldsymbol{d}) = L_0 \int_A W\big(\bar{I}_1\big)\ \mathrm{d}A \approx L_0\, r^2 \sum_k w_k W\big(\bar{I}_1(y_k, z_k)\big)\,, \tag{12}$$

evaluated with a polar quadrature: 4 Gauss–Legendre rings ($\int_0^1 f(s)\, s\, \mathrm{d}s$) $\times$ 16 angles. U_{quad} is the same integral with the small-strain limit $W \to C_{10}(3\varepsilon^2 + \gamma^2)$. The long-term resultants are

$$\boldsymbol{f}_\infty(\boldsymbol{d}) = \boldsymbol{f}_{\text{lin}}(\boldsymbol{d}) + \frac{\partial}{\partial \boldsymbol{d}}\Big(U_{\text{fib}}(\boldsymbol{d}) - U_{\text{quad}}(\boldsymbol{d})\Big), \tag{13}$$

with $\boldsymbol{f}_{\text{lin}}$ from (9) using E_∞, G_∞. At small strain the correction vanishes and the validated linear Timoshenko element is recovered (including its shear flexibility); at large strain axial, bending and torsional responses follow Yeoh, including the net axial force that pure bending produces because of the tension/compression asymmetry. Transverse shear remains linear. The law is conservative. This is the corotational fibre approach (e.g. [1]) with a hyperelastic fibre law; see also geometrically exact beams [2].

5.2 Quasi-linear (Fung) viscoelasticity

With $\beta_i = g_i/(1 - \sum_j g_j)$, the total resultants are $\boldsymbol{f}_{\text{loc}} = \boldsymbol{f}_\infty + \sum_i \boldsymbol{q}_i$, and each Prony branch is driven by increments of the long-term resultants over a step of length h:

$$\boldsymbol{q}_i^{n+1} = a_i\, \boldsymbol{q}_i^n + \beta_i\, b_i\, \big(\boldsymbol{f}_\infty^{n+1} - \boldsymbol{f}_\infty^n\big)\,, \qquad a_i = e^{-h/\tau_i}, \quad b_i = \frac{1 - a_i}{h/\tau_i} = -\frac{\operatorname{expm1}(-h/\tau_i)}{h/\tau_i}. \tag{14}$$

This is exact for a constant rate within the step. For $h \to 0$, $b_i \to 1$ and the response is instantaneous, $\boldsymbol{f}_{\text{loc}} = (E_0/E_\infty)\boldsymbol{f}_\infty$ to first order, which is the stiffness the stable time step must use. The branch histories $(\boldsymbol{q}_i, \boldsymbol{f}_\infty^n)$ are part of the state.

6 Inertia

Lumped translational mass per element end $\rho A L_0/2$. Rotary inertia per element end (scalar, isotropic)

$$j_e = \frac{\rho L_0}{2} \max\!\left(I_y + I_z,\ \frac{A L_0^2}{12}\right), \tag{15}$$

the standard explicit rotary-inertia scaling (it only activates for $L_0 > \sqrt{6}\, r$, i.e. not on the 1 mm results mesh). With isotropic nodal rotary inertia the rotational equation is $j\,\dot{\boldsymbol{\omega}} = \boldsymbol{m}$ with no gyroscopic term.

7 Test rig and boundary conditions

The rig reproduces a drop-weight test on a coupon bonded between two plates:

- **Base plate (bonded, rigid, fixed).** All nodes on the bottom face ($z = 0$) have all six degrees of freedom fixed.
- **Top plate (bonded, rigid).** All nodes on the top face are tied into one rigid body that translates (rotations fixed). Its mass is the plate mass (2 mm aluminium, 7.8 g) plus the lumped mass of the tied nodes. The tie is imposed on accelerations: for tied set $\mathcal{T}$, $\boldsymbol{a}_a \leftarrow \sum_{b\in\mathcal{T}} m_b \boldsymbol{a}_b / \sum_{b\in\mathcal{T}} m_b$ for all $a \in \mathcal{T}$, so the plate moves under the resultant of all forces on its nodes.
- **Impactor.** A rigid body with one translational degree of freedom (z), a separate node without elements, interacting with the top plate only through contact (Section 8.1).
- **Loading: co-moving start.** At $t = 0$ the impactor and the top plate move together at $v_0 = 6\,\mathrm{m/s}$ towards the base, with an impactor–plate gap of 0.1 mm, and the impactor mass is set so that the total kinetic energy equals the impact energy E_{imp} (0.75 J):

$$M = \frac{2E_{\mathrm{imp}}}{v_0^2} - m_{\mathrm{plate}}. \tag{16}$$

  This takes the (plastic) impactor–plate collision as completed, i.e. the plate acts as the striker face. The contact force then builds up continuously as the lattice decelerates the plate and releases continuously on rebound. (Starting from rest with a gap gives a rigid–rigid collision, a force spike and rattling of the light plate; this is physical but not what is measured.)
- **Crush mode.** For constant-velocity crushing (the standard specific-energy test) the impactor mass is set to 1 t, so the plate keeps v_0 independently of the lattice; the run stops at the target stroke.
- Gravity is neglected ($\Delta v = g\,t \approx 0.1\,\mathrm{m/s}$ over 10 ms).

8 Contact and friction

All normal contact laws use the smooth penetration

$$\delta(g) = \varepsilon\ \mathrm{softplus}(-g/\varepsilon), \tag{17}$$

which tends to $\max(-g, 0)$ for $|g| \gg \varepsilon$ and is C^∞. Unless stated otherwise $\varepsilon = 0.05\,r_{\min}$. Under light load contact therefore acts within $\approx 2\varepsilon$ of geometric touching (the regularisation width).

8.1 Impactor–plate contact: critically damped Hertz law

With gap $g = g_0 + u_{z,\mathrm{imp}} - u_{z,\mathrm{plate}}$ and gap rate $\dot{g}$, the contact force on the impactor is

$$f = k\,\delta^{3/2} - c(\delta)\,\dot{g}, \qquad c(\delta) = 2\zeta\sqrt{k_t\,\mu_r}, \quad k_t = \tfrac{3}{2}k\,\delta^{1/2}, \quad \mu_r = \frac{M\,m_{\mathrm{plate}}}{M + m_{\mathrm{plate}}}, \quad \zeta = 1, \tag{18}$$

followed by a smooth positive part that is exactly zero at $f = 0$:

$$F = \max\Big(0,\ \tfrac{1}{2}\left(f + \sqrt{f^2 + s^2}\right) - \tfrac{1}{2}s\Big), \qquad s = 0.01\,k\,\delta^{3/2}. \tag{19}$$

k is calibrated so that the law carries $F_{\mathrm{ref}} = 2000\,\mathrm{N}$ at $\delta_{\mathrm{ref}} = 0.2\,\mathrm{mm}$ ($k = F_{\mathrm{ref}}/\delta_{\mathrm{ref}}^{3/2}$, $\varepsilon = 0.1\,\delta_{\mathrm{ref}}$). The damping, tuned to the impactor–plate contact mode itself, removes the ringing of the light plate between the heavy impactor and the soft lattice. It vanishes as $\delta^{1/4}$ at onset (continuous force), and the positive part prevents sticky tension at separation. Alternatives implemented for comparison: linear penalty, Hertz, Hunt–Crossley [7, 8] and an IPC log-barrier [9].

8.2 Beam–beam contact: line-contact density

Every element is a capsule: its centreline segment with radius r_e. Rather than one point force per pair of segments, contact is a *density integrated along the beams*.

Quadrature points. Each element a carries $n_q = \lceil L_{\max}/h_q \rceil$ points $\boldsymbol{p}_{a,k} = \boldsymbol{x}_{a_1} + \xi_k(\boldsymbol{x}_{a_2} - \boldsymbol{x}_{a_1})$, $\xi_k = (k + \frac{1}{2})/n_q$, with tributary length $w_a = L_{0,a}/n_q$ ($h_q = 0.1\,\mathrm{mm}$).

Gap to another beam. For a point $\boldsymbol{p}$ and an element b with end points $\boldsymbol{q}_0$, $\boldsymbol{q}_0 + \boldsymbol{d}_b$, the projection is $t = \mathrm{clamp}\,((\boldsymbol{p} - \boldsymbol{q}_0) \cdot \boldsymbol{d}_b/(\boldsymbol{d}_b \cdot \boldsymbol{d}_b), 0, 1)$, and the gap of point (a, k) to beam B (all candidate elements of B) is the distance to the polyline minus both radii,

$$g^B_{a,k} = \min_{b \in B} \left\| \boldsymbol{p}_{a,k} - \boldsymbol{q}^b_0 - t\,\boldsymbol{d}_b \right\| - r_a - r_B. \tag{20}$$

Taking the minimum over the elements of B (rather than summing over element pairs) avoids counting a contact twice where two elements of B share a node. t is kept in the derivative: the force (first derivative) would be exact with t frozen (envelope theorem), but the contact *stiffness*, needed for trajectory sensitivities, contains $\partial t/\partial \boldsymbol{x}$.

Potential.

$$\Phi_{\mathrm{bb}} = \frac{1}{2} \sum_{a,k} \sum_{B \neq \mathrm{beam}(a)} w_a \, \frac{k_\ell}{2} \, \delta\big(g^B_{a,k}\big)^2, \qquad k_\ell = \frac{E^*}{2}, \quad E^* = \frac{E_0}{2(1-\nu^2)}, \tag{21}$$

and the nodal forces are $-\partial \Phi_{\mathrm{bb}}/\partial \boldsymbol{x}$ (automatic differentiation), so the contact is conservative. The factor $\frac{1}{2}$ accounts for each contact zone being integrated from both beams.

Line and point contact from one law. Two parallel beams overlapping over a length ℓ carry $F = k_\ell\,\delta\,\ell$ (line contact). For two crossed cylinders of radius R the capsule gap along one beam is $g(x) \approx -\delta + x^2/(4R)$, and integrating the density over the contact zone gives

$$F = \int k_\ell \left(\delta - \frac{x^2}{4R} \right)_+ \mathrm{d}x = \frac{4}{3}\, k_\ell \sqrt{4R}\, \delta^{3/2} = \frac{4}{3} E^* \sqrt{R}\, \delta^{3/2} \quad \text{for } k_\ell = E^*/2, \tag{22}$$

i.e. exactly the Hertz law for crossed cylinders [6], without using it. (Numerically: 0.99, 0.98, 0.97 of Hertz at $\delta = 0.02, 0.05, 0.1\,\mathrm{mm}$.)

Exclusions (geometric, mesh independent).

1. A beam does not contact itself.
2. A (point, beam) combination already overlapping in the reference configuration ($g \leq 0$) never contacts.
3. Joint crotch: if the two beams share a lattice node, a (point, beam) combination whose reference gap is below one radius is treated as joint material. Otherwise any joint rotation drives the crotch surfaces into "contact" and double counts the joint stiffness, which the beam element already provides.

Contact between struts of the same joint further out (folding struts) remains active.

Candidate search. Element pairs from different beams with segment–segment gap below a margin (1 mm) are kept in a Verlet list. It is refreshed when any node has moved more than half the margin since the last refresh, so no pair can close unseen. For each step only the pairs whose segment gap is below 6ε are evaluated point by point; beyond this the potential is below e^{-12} of the contact scale. Both choices are deterministic functions of the state (needed for the adjoint).

8.3 Beam–plate contact

Every non-bonded lattice node a contacts both plates with a Hertz potential of a sphere of the nodal radius r_a (largest radius of the attached elements) on a rigid flat:

$$\Phi_{\mathrm{pl}} = \sum_a \frac{2}{5} k_a\, \delta(g_a)^{5/2}, \qquad k_a = \frac{4}{3}\frac{E_0}{1-\nu^2}\sqrt{r_a}, \qquad g_a^{\mathrm{base}} = z_a - r_a, \quad g_a^{\mathrm{top}} = z_{\mathrm{plate}} - z_a - r_a. \tag{23}$$

The reaction on the top plate enters the plate's equation of motion through z_{plate}.

8.4 Friction

Coulomb friction with $\mu = 0.5$ acts at every contacting line-contact point and plate contact. With normal force F_n (from the same potential) and tangential slip velocity $\boldsymbol{v}_t$ between the two contacting material points (the point on beam a and its closest point on beam b, or node and plate),

$$\boldsymbol{F}_t = -\mu F_n \frac{\boldsymbol{v}_t}{\sqrt{\|\boldsymbol{v}_t\|^2 + v_\varepsilon^2}}, \qquad v_\varepsilon = \max\!\left(v_{\varepsilon 0},\ \frac{2\mu F_n h}{m_{\mathrm{eff}}}\right), \quad v_{\varepsilon 0} = 1\,\mathrm{mm/s}, \tag{24}$$

with $m_{\mathrm{eff}} = m_p m_q/(m_p + m_q)$ the reduced mass of the two contact points (masses interpolated from the element nodes). The equivalent viscous coefficient $\mu F_n/v_\varepsilon$ never exceeds $m_{\mathrm{eff}}/(2h)$, so near stick the explicit step stays stable and the law acts as a stick constraint for slip speeds below $\approx \mu F_n h/m_{\mathrm{eff}}$; under real slip it is Coulomb's μF_n. The forces are applied with the interpolation weights at both contact points (equal and opposite).

9 Explicit time integration

9.1 Velocity Verlet with exponential-map rotations

With nodal accelerations $\boldsymbol{a} = (\boldsymbol{f}_{\mathrm{ext}} + \boldsymbol{f}_{\mathrm{c}} - \boldsymbol{f}_u)/m$ and angular accelerations $\boldsymbol{\alpha} = -\boldsymbol{f}_\theta/j$ (fixed degrees of freedom masked), one step of length h_n is

$$\boldsymbol{v}^{n+1/2} = \boldsymbol{v}^n + \tfrac{h_n}{2}\boldsymbol{a}^n, \qquad \boldsymbol{\omega}^{n+1/2} = \boldsymbol{\omega}^n + \tfrac{h_n}{2}\boldsymbol{\alpha}^n, \tag{25}$$

$$\boldsymbol{u}^{n+1} = \boldsymbol{u}^n + h_n \boldsymbol{v}^{n+1/2}, \qquad \mathbf{R}^{n+1} = \exp\!\Big([\, h_n \boldsymbol{\omega}^{n+1/2}\,]_\times\Big)\,\mathbf{R}^n, \tag{26}$$

$$\boldsymbol{a}^{n+1},\ \boldsymbol{\alpha}^{n+1} = \text{forces at } (\boldsymbol{u}^{n+1}, \mathbf{R}^{n+1}, \boldsymbol{v}^{n+1/2}), \tag{27}$$

$$\boldsymbol{v}^{n+1} = \boldsymbol{v}^{n+1/2} + \tfrac{h_n}{2}\boldsymbol{a}^{n+1}, \qquad \boldsymbol{\omega}^{n+1} = \boldsymbol{\omega}^{n+1/2} + \tfrac{h_n}{2}\boldsymbol{\alpha}^{n+1}, \tag{28}$$

followed by the Prony update (14). The exponential map is evaluated with the Rodrigues formula in a form that is smooth at zero rotation, $\exp([\,\boldsymbol{w}\,]_\times) = \mathbf{I} + \mathrm{sinc}(r)[\,\boldsymbol{w}\,]_\times + \frac{1}{2}\,\mathrm{sinc}(r/2)^2[\,\boldsymbol{w}\,]_\times^2$, $r = \|\boldsymbol{w}\|$. One force evaluation per step; velocity-dependent forces (damping, friction) see the half-step velocity.

9.2 Stable time step

The step is adapted every step as

$$h_n = \min\!\left(\frac{2C}{\sqrt{\omega_{\mathrm{beam}}^2 + \omega_{\mathrm{c}}^2}},\ h_{\mathrm{cap}}\right), \qquad C = 0.7, \tag{29}$$

where:

- $\omega_{\mathrm{beam}}^2 = \lambda_{\max}(\mathbf{M}^{-1/2}\mathbf{K}\mathbf{M}^{-1/2})$ of the current tangent is estimated by a warm-started power iteration (Jacobian–vector products of the internal force, free–free, i.e. an upper bound for any support), refreshed every 20 steps. With $h \ll \tau_1$ the Prony law makes the tangent the instantaneous (E_0) one.

- ω_c collects contact stiffnesses. For the impactor law, $\omega = \sqrt{k_t/\mu_r}/(\sqrt{1+\zeta^2}-\zeta)$ (damped central-difference limit). For line and plate contact, every quadrature point's exact stiffness $\frac{1}{2} w_a k_\ell (\sigma^2 + \delta\sigma(1-\sigma)/\varepsilon)$ is summed per element and per node (a row-sum bound, which covers nodes in several contacts at once), giving $\sqrt{2K_{\text{node}}/m_{\text{node}}}$.
- h_{cap} is a kinematic cap: near contact, the relative approach per step is limited to 0.1ℓ ($\ell = \delta_{\text{ref}}$ for the impactor law, 2ε for lattice contact), so the force cannot jump from zero to a large value within one step. The stiffness-based estimate cannot see this, because the stiffness is zero just before contact.

If h_n falls below 10^{-3} of its initial value the run stops (instability guard). Typical values: $h \approx 4 \times 10^{-6}\,\text{s}$ on the 1 mm mesh with E_0; an 8 ms drop takes ≈ 7400 steps.

10 Energy accounting

The energy balance is monitored as

$$E_{\text{tot}} = \underbrace{\tfrac{1}{2}\sum m\|\boldsymbol{v}\|^2 + \tfrac{1}{2}\sum j\|\boldsymbol{\omega}\|^2}_{\text{kinetic}} + \underbrace{\sum_n \tfrac{1}{2}(\boldsymbol{f}_{\text{loc}}^n + \boldsymbol{f}_{\text{loc}}^{n+1})\cdot(\boldsymbol{d}^{n+1} - \boldsymbol{d}^n)}_{\text{internal work (stored + dissipated)}} + \Phi_{\text{imp}} + W_{\text{damp,imp}} + \Phi_{\text{bb}} + \Phi_{\text{pl}} + W_{\text{fric}}, \quad (30)$$

where $W_{\text{fric}} = -\sum_n \frac{1}{2}(\boldsymbol{F}_t^n + \boldsymbol{F}_t^{n+1})\cdot(\boldsymbol{u}^{n+1} - \boldsymbol{u}^n)$ is the work of the friction forces actually applied. The material dissipation is the internal work minus the stored long-term energy (fibre Yeoh plus quadratic parts).

11 Response measures, objectives and exact sensitivities

11.1 Response curve and objectives

The response is the impactor force F versus top-plate stroke δ. Per step, $\Delta\delta = \delta^{n+1} - \delta^n$, $\bar{F} = \frac{1}{2}(F^n + F^{n+1})$, $\bar{\delta} = \frac{1}{2}(\delta^n + \delta^{n+1})$, and

$$S_0 = \sum_n \big(\bar{F} - F^*(\bar{\delta})\big)^2\,\text{relu}(\Delta\delta), \qquad S_1 = \sum_n \bar{F}\;\text{relu}(\Delta\delta), \qquad S_2 = \sum_n \bar{F}\,\Delta\delta. \quad (31)$$

- **Target force–displacement curve (inverse design):** $J_{\text{MSE}} = S_0 / \int F^{*2}\,\mathrm{d}\delta$, with F^* piecewise linear (e.g. an "ideal absorber" plateau [14]).
- **Specific energy absorption:** $\text{SEA} = S_k/m_{\text{lat}}$ with $m_{\text{lat}} = \rho\sum AL_0$; $k = 1$ (loading area) for constant-velocity crushing, $k = 2$ (dissipated loop area) in a fixed-energy drop. There the loading area is close to the impact energy whatever the design, so the loading-area SEA would reduce to $1/m$. The objective $J = -\text{SEA}$ is minimised.

11.2 Discrete adjoint with two-level checkpointing

Write one time step as the pure map $\boldsymbol{z}^{n+1} = \Phi_n(\boldsymbol{z}^n; \boldsymbol{\theta})$ with state $\boldsymbol{z} = (\boldsymbol{u}, \boldsymbol{v}, \mathbf{R}, \boldsymbol{\omega}, \boldsymbol{a}, \boldsymbol{\alpha}, \boldsymbol{q}, \boldsymbol{f}_\infty, F_{\text{b}}, F)$, and the objective as $J = g(\boldsymbol{S}, \boldsymbol{\theta})$ with additive accumulators $\boldsymbol{S} = \sum_n \boldsymbol{\ell}_n(\boldsymbol{z}^n, \boldsymbol{z}^{n+1})$. The time-step schedule $\{h_n\}$ of the nominal run is replayed, and design-dependent rig constants (impactor mass

and law, ε, exclusion sets, plate contact node sets) are frozen at the nominal design. The contact Verlet list is stored with each checkpoint and restored on recompute. The adjoint recursion is

$$\boldsymbol{\lambda}^n = \left(\frac{\partial \Phi_n}{\partial \boldsymbol{z}^n}\right)^\top \boldsymbol{\lambda}^{n+1} + \frac{\partial(\boldsymbol{w}\cdot\boldsymbol{\ell}_n)}{\partial \boldsymbol{z}^n}, \qquad \frac{\mathrm{d}J}{\mathrm{d}\boldsymbol{\theta}} = \frac{\partial g}{\partial \boldsymbol{\theta}} + \sum_n \left[\left(\frac{\partial \Phi_n}{\partial \boldsymbol{\theta}}\right)^\top \boldsymbol{\lambda}^{n+1} + \frac{\partial(\boldsymbol{w}\cdot\boldsymbol{\ell}_n)}{\partial \boldsymbol{\theta}}\right], \qquad \boldsymbol{w} = \frac{\partial g}{\partial \boldsymbol{S}}, \tag{32}$$

with each product a single-step vector–Jacobian product.

Algorithm 1 Checkpointed discrete adjoint (`adjoint.checkpointed_adjoint`)

forward (no graph): $\boldsymbol{z}^0$; for $n = 0..N-1$: store $\boldsymbol{z}^n$ and contact state if $n \bmod K = 0$; $\boldsymbol{z}^{n+1} = \Phi_n(\boldsymbol{z}^n)$, $\boldsymbol{S}$ += $\boldsymbol{\ell}_n$
$\boldsymbol{w} \leftarrow \partial g/\partial \boldsymbol{S}$, $\boldsymbol{g}_\theta \leftarrow \partial g/\partial \boldsymbol{\theta}$, $\boldsymbol{\lambda} \leftarrow 0$
for segments $[s, s+K)$ from last to first **do**
    restore checkpoint s; recompute and store $\boldsymbol{z}^{s..s+K-1}$ (no graph)
    for $n = s+K-1$ down to s **do**
        build the graph of one step $\boldsymbol{z}^{n+1}, \boldsymbol{\ell}_n = \Phi_n(\boldsymbol{z}^n)$ from the stored $\boldsymbol{z}^n$
        $(\boldsymbol{\lambda}, \Delta\boldsymbol{g}_\theta) \leftarrow \mathrm{VJP}\big((\boldsymbol{z}^{n+1}, \boldsymbol{w}\cdot\boldsymbol{\ell}_n);\ (\boldsymbol{\lambda}, 1)\big)$; $\boldsymbol{g}_\theta$ += $\Delta\boldsymbol{g}_\theta$
    end for
end for
chain $\boldsymbol{\lambda}^0$ through $\boldsymbol{z}^0(\boldsymbol{\theta})$

Memory is $\mathcal{O}((N/K + K)\,|\boldsymbol{z}|)$ plus one step graph, $K = \sqrt{N}$, independent of the number of design variables. The gradient equals reverse-mode differentiation of the unrolled loop (discretise-then-differentiate [13]); checkpointing follows [12]. Measured on the 3 mm mesh at 1 ms: unrolled reverse AD 327 s and 16.7 GB (CPU), adjoint 29.8 s and 0.69 GB; the full 8 ms drop on the 1 mm mesh takes 403 s and 0.43 GB on an 8 GB GPU.

11.3 Optimisers

Design variables are normalised to $[0, 1]$ and denormalised inside the objective. For many design variables with constraints the method of moving asymptotes (MMA, [10]) is used with asymptotes carried between iterations, or its globally convergent variant GCMMA [11], whose inner loop enlarges the approximation curvature using value-only (forward) evaluations until the approximation is conservative. L-BFGS-B is used for small unconstrained problems.

12 Verification

check	result
small-strain tangent vs. classical Timoshenko stiffness	equal to 10^{-9} (skewed and bent frames, $I_y \neq I_z$)
rigid rotations up to 172°	internal force $< 10^{-10}$
geometric stiffness: column under axial load P	$\omega_1^2(P)/\omega_1^2(0) = 1 - P/P_{\mathrm{cr}}$ within 2 % (Engesser P_{cr})
cantilever first frequency	within 0.12 % of Euler–Bernoulli
tip moment $2\pi EI/L$	beam closes to a full circle (tip error 1.4×10^{-5} mm)
fibre Yeoh: bending moment, torque	within 0.1 % / 1 % of direct section integration
Prony: step relaxation; loss per cycle at 1 kHz	exact $E_\infty + (E_0 - E_\infty)e^{-t/\tau}$; $\pi E'' \varepsilon_0^2 V$ within 2 %
line contact, crossed cylinders	Hertz within 1–3 % (from integration alone)

beam radius in contact	no force for $d > r_A + r_B$; onset at $r_A + r_B$ (radii 0.8/0.5)
two free beams colliding	energy within 1 %, momentum to 10^{-9}, no interpenetration
friction	$\|\boldsymbol{F}_t\| = \mu F_n$ under slip; power $= \mu F_n \|\boldsymbol{v}\|$; equal and opposite
drop energy balance	$\leq 1.1 \times 10^{-4}$ relative (all physics)
contact force continuity	per-step force change $\propto h$ (no jump)
mesh convergence (0.75 vs 1.0 mm)	stroke 2 %, dissipation 0.2 %, densification peak ± 3.5 %
adjoint vs. unrolled AD; vs. finite differences	10^{-16}; 10^{-9}–10^{-10} (non-uniform designs, all physics)

13 Limitations

- One Prony term ($\tau_1 = 52\,\mu$s): most of the dissipation occurs within ≈ 1 ms, and the slow viscoelasticity near T_g is absent. The relaxation spectrum is the most influential uncertain input.
- Transverse shear is linear; only the axial, bending and torsional fibre responses are hyperelastic.
- Joints are centreline intersections: overlapping material is double counted in the mass, and joint stiffness comes from the beam ends only.
- Contact acts within $\approx 2\varepsilon$ of touching under light load (regularisation width $\varepsilon = 0.05\, r_{\min}$).
- The co-moving start assumes a completed plastic impactor–plate collision.
- Friction mostly sticks at these loads and dissipates < 1 % of the impact energy.

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